101 · RESEARCH UTILITIES

Hi-fi explorations

VSCode-grammar research workspace · amber CRT aesthetic · IBM Plex Mono throughout. Two layout directions, seven artboards.

LAYOUT A · REFINED IDE
4 artboards · pages 2–5

LAYOUT B · CONSOLE
3 artboards · pages 6–8

101 / vault / transformers / attention

⌕ search vault, run command, ask… ⌘K

● sync 234 papers @gengeltje

101

VAULT V · FIND F · READ R · GRAPH G · ASK A · STUDY S

EXPLORER

/ filter vault… ⌘P

PINS

- ◷ Recent 12 [1]
- Reading list 8 [2]
- ★ Starred 14 [3]
- ▾ Saved searches 5
  - ssl + dino + frontier 8
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VAULT · 234 PAPERS

- ▾ transformers 58 [T]
  - attention 22
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PROJECTS

- ◆ thesis · vision-ssl 47
- ◆ reading-group / w12 6

TAGS

- # foundational 18
- # frontier 31
- # to-implement 9
- # needs-review 14

◆ /transformers/attention ● ×

⇆ split ⊟ layout

FOLDER vault / transformers / attention

attention

22 papers · 3 unread · last add 2d

＋ ADD | ↑ IMPORT | ↓ EXPORT .BIB | ⋯

foundational ×8 | transformer ×22 | attention ×22 | NeurIPS ×6 | ICLR ×4 | 2017–2024

filter [/] sort [S] · recent view [⌘1] · li:

● PAPERS 22 ○ NOTES 14 ○ ANNOTATIONS 87 ○ GRAPH ○ Q&A [1] [2] [3] [4] [5]

YEAR	VENUE	TITLE	AUTHORS	CITES	NOTE	ANN	STATUS
2017	NeurIPS	Attention Is All You Need	A. Vaswani, N. Shaze…	134.8k	◆	↵ READ ·	O OPEN
2021	ICCV	Emerging Properties in Self-Supervised Vision…	M. Caron, H. Touvron…	8.8k	◆	↵ READ ·	O OPEN
2021	ICLR	An Image is Worth 16x16 Words: Transformers f…	A. Dosovitskiy, et a…	41.2k	◆3	11	READ
2019	NAACL	BERT: Pre-training of Deep Bidirectional Tran…	J. Devlin, M.-W. Cha…	102.0k	◆1	6	READ
2019	OpenAI Tech Report	Language Models are Unsupervised Multitask Le…	A. Radford, et al.	14.8k	·	·	READ
2022	CVPR	Masked Autoencoders Are Scalable Vision Learn…	K. He, X. Chen +2	7.0k	◆2	8	READ
2020	ICML	A Simple Framework for Contrastive Learning o…	T. Chen, S. Kornblit…	18.5k	·	·	READ
2020	NeurIPS	Bootstrap Your Own Latent: A New Approach to …	J.-B. Grill, et al.	6.3k	·	·	READ
2024	TMLR	DINOv2: Learning Robust Visual Features witho…	M. Oquab, T. Darcet …	1.8k	◆1	·	READING
2023	CVPR	Self-Supervised Learning from Images with a J…	M. Assran, et al.	612	·	·	UNREAD
2022	ACM Computing Surveys	Efficient Transformers: A Survey	Y. Tay, M. Dehghani …	1.5k	·	·	UNREAD
2020	arXiv	Scaling Laws for Neural Language Models	J. Kaplan, et al.	5.8k	·	·	READ

INSPECTOR

● attention

folder · 22 papers

READING STATS

- read 18 / 22
- reading 2 / 22
- unread 2 / 22

AUTHORS · TOP CO-OCCURRING

- A. Vaswani ×6
- N. Shazeer ×4
- J. Devlin ×3
- Y. Bengio ×3
- K. He ×2
- M. Caron ×2

ASK THIS FOLDER

What are the main arguments against scaled dot-product attention's $O(n^2)$ complexity?

RUN ↵ cite 22

RECENTLY OPENED

- ▸ Vaswani 2017 — 2h
- ▸ Devlin 2019 — yesterday
- ▸ Tay 2022 — 3d

101

VAULT V

FIND F

READ R

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ASK A

STUDY S

EXPLORER

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?

/transformers/attention × | Vaswani 2017 · Attention… ● ×

split layout

PAPER arXiv:1706.03762 · NeurIPS 2017 foundational transformer

cited 134.8k · in vault since 2024-04-12

Attention Is All You Need

Vaswani · Shazeer · Parmar · Uszkoreit · Jones · Gomez · Kaiser · Polosukhin

TEXT [T] PDF [P] SPLIT [S] ✎ NOTES ⌥ ANNOTATE ◇ GRAPH + ASK

[H] highlight · [N] note · [Q] question · [F] flag

inherently sequential nature precludes parallelization within training examples, which becomes critical at longer sequence lengths, as memory constraints limit batching across examples.

◆ note · me · 2d

¶4 Attention mechanisms have become an integral part of compelling sequence modeling and transduction models in various tasks, allowing modeling of dependencies without regard to their distance in the input or output sequences. In all but a few cases, however, such attention mechanisms are used in conjunction with a recurrent network.

¶5 In this work we propose the Transformer, a model architecture eschewing recurrence and instead relying entirely on an attention mechanism to draw global dependencies between input and output. The Transformer allows for significantly more parallelization and can reach a new state of the art in translation quality after being trained for as little as twelve hours on eight P100 GPUs.

? question · me · 2d

2 Background

¶7 The goal of reducing sequential computation also forms the foundation of the Extended Neural GPU, ByteNet and ConvS2S, all of which use convolutional neural networks as basic building block, computing hidden representations in parallel for all input and output positions. In these models, the number of operations required to relate signals from two arbitrary input or output positions grows in the distance between positions, linearly for ConvS2S and logarithmically for ByteNet.

¶8 Self-attention, sometimes called intra-attention, is an attention mechanism relating different positions of a single sequence in order to compute a representation of the sequence. Self-attention has been used successfully in a variety of tasks including reading comprehension, abstractive summarization, textual entailment and learning task-independent sentence representations.

— PAGE 2 OF 15 —

◄ p.2 / 15 ► scroll · ↑↓ · or [j k]

selection · 0 chars [H] highlight [N] note [Q] question [⌘↵] ask

MARGIN · §1

◆ NOTE · ¶3 · me · 2d

Key motivation — RNN parallelism wall.
See Chen 2018 §3 for follow-up.

? QUESTION · ¶5 · me · 2d

Q: how does this compare to ByteNet / ConvS2S in §2?

[A] ask vault [L] link paper

◆ HIGHLIGHT · ¶8 · me · today

"Self-attention … relating different positions of a single sequence" → tie to DINO's CLS-token attention maps.

INSPECTOR

Vaswani 2017

paper · ¶ 3 · selected

LINEAGE 3 ← · → 5

CITES ←

- Bahdanau 2014 soft alignment seed
- Luong 2015 dot-prod variant
- Cheng 2016 intra-attention

CITED BY →

- Devlin 2019 BERT
- Radford 2019 GPT-2
- Dosovitskiy 2021 ViT
- Brown 2020 GPT-3
- Touvron 2023 LLaMA

OPEN GRAPH ◇

ASK THIS PAPER

YOU

Explain scaled dot-product attention vs additive attention.

101 · cites ¶8, eq.(1), §3.2.1

Scaled dot-product computes $Q·K^T / \sqrt{d_k}$ and softmaxes [¶8]. Additive uses a feed-forward net with a single hidden layer [§3.2.1]. The authors find dot-product faster in practice…

↵ + follow-up 3 / 8 left

METADATA

doi 10.48550/arXiv.1706.03762
venue NeurIPS 2017 · Long Beach
pages 15 · 5,084 words
cite vaswani2017attention

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/transformers/attention × | Vaswani 2017 · Attention… ● × | split layout

PAPER arXiv:1706.03762 · NeurIPS 2017 foundational transformer | cited 134.8k · in vault since 2024-04-12

Attention Is All You Need

Vaswani · Shazeer · Parmar · Uszkoreit · Jones · Gomez · Kaiser · Polosukhin

TEXT T | PDF P | SPLIT S | ✎ NOTES | ANNOTATE | ◇ GRAPH | + ASK | [H] highlight · [N] note · [Q] question · [F] flag

Ashish Vaswani · Noam Shazeer · Niki Parmar · Jakob Uszkoreit
Llion Jones · Aidan N. Gomez · Łukasz Kaiser · Illia Polosukhin
Google Brain · Google Research · University of Toronto

Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely.

1 Introduction

Recurrent neural networks, long short-term memory and gated recurrent neural networks in particular, have been firmly established as state of the art approaches in sequence modeling and transduction problems such as language modeling and machine translation.

Numerous efforts have since continued to push the boundaries of recurrent language models and encoder-decoder architectures.

Recurrent models typically factor computation along the symbol positions of the input and output sequences. Aligning the positions to steps in computation time, they generate a sequence of hidden states h_t, as a function of the previous hidden state h_{t-1}.

This inherently sequential nature precludes parallelization within training examples, which becomes critical at longer sequence lengths, as memory constraints limit batching across examples.

Attention mechanisms have become an integral part of compelling sequence modeling and transduction models in various tasks, allowing modeling of dependencies without regard to their distance in the input or output sequences.

In this work we propose the Transformer, a model architecture eschewing recurrence and instead relying entirely on an attention mechanism to draw global dependencies between input and output.

◄ p.2 / 15 ► zoom 100% · ⌘0 fit | PDF · text layer ON · render via pdf.js | T text view

INSPECTOR

Vaswani 2017

paper · ¶ 3 · selected

LINEAGE 3 ← · → 5

CITES ←

- Bahdanau 2014 soft alignment seed
- Luong 2015 dot-prod variant
- Cheng 2016 intra-attention

CITED BY →

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- Touvron 2023 LLaMA

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/transformers/attention × | Vaswani 2017 × | Discover · ssl + dino ● × | split | layout

DISCOVER · SEED-BASED

type to refine seed · chips combine with AND

★ Saved as ssl + dino + frontier · in /Pins/Saved searches ✎ | ↺ rerank | +

:seed | @ caron2021 DINO × | / self-supervised 41p × | # frontier × | year:>2022 × | caron

↑↓ pick · ⏎ add · ⌫ remove

@	**Vision Transformers Need Registers** Darcet 2024 · ICLR	cit 412
@	**I-JEPA** Assran 2023 · CVPR	cit 612
@	**DINOv2** Oquab 2024 · TMLR · ★ vault	cit 1.8k

FOLDERS — 4 match

/	**interpretability** vault folder	28 papers
/	**vision-transformers** vault folder	33 papers

TAGS — 3 match

#	**jepa** tag	6 papers
#	**masked** tag	11 papers

AUTHORS — 2 match

☻	**Mathilde Caron** Meta AI · 7 in vault	
☻	**Yann LeCun** Meta AI · 3 in vault	

SCOPE: @paper · /folder · #tag · author:name · venue:NeurIPS · year:>2022 esc

↳ Often co-cited with ViT (Dosovitskiy 2021) — vit, convnext — PREVIEW · GRAPH

0.86 Scaling Vision Transformers to 22 Billion Parameters
M. Dehghani, et al. · ICML · 2023 · cited 423
↳ Extends a paper you starred · scaling-laws folder — vit, scaling, frontier
+ ADD · PREVIEW · GRAPH

0.83 Masked Siamese Networks for Label-Efficient Learning
M. Assran, et al. · ECCV · 2022 · cited 514
↳ Bridges your contrastive and masked-modeling folders — ssl, contrastive, masked
+ ADD · PREVIEW · GRAPH

0.99 Emerging Properties in Self-Supervised Vision Transformers IN VAULT
M. Caron, et al. · ICCV · 2021 · cited 8.8k
↳ IN YOUR VAULT · /self-supervised/dino-family — ssl, dino, vit
OPEN

SEARCH CONSOLE

✦ SEARCH CONSOLE
4 seeds · 3 agents · 8 candidates

YEAR RANGE
2018 — 2024

VENUES
NeurIPS · ICML · ICLR · CVPR · ICCV · ECCV · TMLR · arXiv

SIGNAL MIX

semantic	80
co-cite	65
author net	40
novelty	55
venue tier	30

drag to retune ranking

AGENTS · PARALLEL SEARCH
1 2 3 4 5 6 7 8
● semantic ⓘ ● citation ⓘ ● author ⓘ · +5 idle

SCHEDULE

- On vault add — incremental — last run 12m ago
- Daily 09:00 — full re-rank — next in 6h
- Weekly · Mon — deep search across arXiv
- One-shot — run now once · ⌘⏎

BATCH
+ ADD 5 STARRED | ↓ .bib

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